WORKED SOLUTION

STEP III 2025, Question 3

Pure • Inequalities of Means

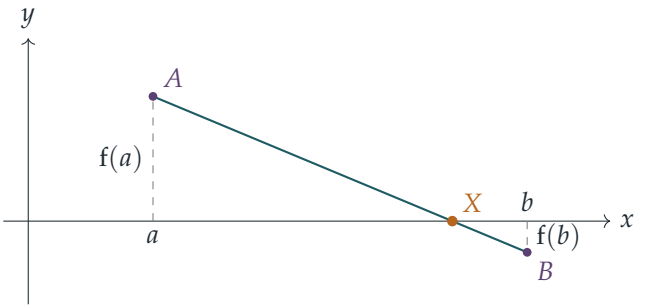
STEP III 2025, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Because A sits above the axis and B below it, the crossing point X divides AB in the ratio of the two heights $\mathrm{f}(a) : \mathrm{f}(b)$. So m is a *weighted average* of a and b, and everything that follows is about how the weights change as f changes. Once m is written as a function of the single ratio $\mathrm{f}(a)/\mathrm{f}(b)$, comparing two choices of f becomes comparing two numbers.



Solution

Part (i)

The triangles formed by A, its foot $(a, 0)$ and X, and by B, its foot $(b, 0)$ and X, are similar (vertically opposite angles at X, right angles at the feet). Hence

$$\frac{m-a}{b-m} = \frac{\mathrm{f}(a)}{\mathrm{f}(b)} \quad \Longrightarrow \quad m\,\mathrm{f}(b) - a\,\mathrm{f}(b) = b\,\mathrm{f}(a) - m\,\mathrm{f}(a),$$

so

$$\boxed{m = \frac{a\,\mathrm{f}(b) + b\,\mathrm{f}(a)}{\mathrm{f}(a) + \mathrm{f}(b)}}.$$

(The denominator is positive because $f > 0$; the same answer comes from the line's equation $y - f(a) = -\frac{f(a)+f(b)}{b-a}(x-a)$ with $y = 0$.)

It is worth rewriting this with $r = f(a)/f(b) > 0$, dividing top and bottom by $f(b)$:

$$m = \frac{a+br}{1+r} = b - \frac{b-a}{1+r}. \qquad (*)$$

Part (ii)

With $f(x) = \sqrt{x}$,

$$m = \frac{a\sqrt{b} + b\sqrt{a}}{\sqrt{a}+\sqrt{b}} = \frac{\sqrt{a}\sqrt{b}\,(\sqrt{a}+\sqrt{b})}{\sqrt{a}+\sqrt{b}} = \sqrt{ab},$$

as required.

For the second target, the numerator $a^{n+1} + b^{n+1}$ should come from $a\,f(b) + b\,f(a)$ after clearing fractions, which suggests $f(b)$ should behave like b^{-n}. Try $f(x) = x^{-n}$, which is positive for $x > 0$:

$$m = \frac{ab^{-n} + ba^{-n}}{a^{-n} + b^{-n}} = \frac{a^{n+1} + b^{n+1}}{b^n + a^n},$$

multiplying top and bottom by $a^n b^n$. So $f(x) = x^{-n}$ works (as does any positive constant multiple of it, since m only depends on the ratio $f(a)/f(b)$). Note that $n = -\frac{1}{2}$ recovers the first result.

Part (iii)

Let $r_1 = g_1(a)/g_1(b)$ and $r_2 = g_2(a)/g_2(b)$, both positive. By $(*)$, $M_i = b - \dfrac{b-a}{1+r_i}$. Since $b - a > 0$, the expression $b - \frac{b-a}{1+r}$ is strictly increasing in r for $r > 0$, so it is enough to show $r_1 > r_2$.

> **Ask yourself** How does a condition on the *function* g_1/g_2 turn into a condition on r_1 and r_2?

Write $h = g_1/g_2$. Then

$$\frac{r_1}{r_2} = \frac{g_1(a)\,g_2(b)}{g_1(b)\,g_2(a)} = \frac{h(a)}{h(b)} > 1,$$

because h is decreasing and $a < b$ give $h(a) > h(b) > 0$. As $r_2 > 0$, this gives $r_1 > r_2$, and hence $M_1 > M_2$.

Hence. We need three functions whose values of m are the three means.

- $f(x) = 1$ (constant) gives $m = \dfrac{a+b}{2}$.
- $f(x) = \sqrt{x}$ gives $m = \sqrt{ab}$, by part (ii).

▶ $\mathrm{f}(x) = x$ gives $m = \dfrac{ab + ba}{a + b} = \dfrac{2ab}{a + b}$.

Now apply the result just proved twice. With $\mathrm{g}_1 = 1$, $\mathrm{g}_2 = \sqrt{x}$, the ratio $\mathrm{g}_1/\mathrm{g}_2 = x^{-1/2}$ is decreasing for $x > 0$, so $\frac{a+b}{2} > \sqrt{ab}$. With $\mathrm{g}_1 = \sqrt{x}$, $\mathrm{g}_2 = x$, the ratio is again $x^{-1/2}$, decreasing, so $\sqrt{ab} > \frac{2ab}{a+b}$. Together,

$$\frac{a + b}{2} > \sqrt{ab} > \frac{2ab}{a + b}.$$

Part (iv)

The curve passes through A and B, so

$$\mathrm{f}(a) = p(c - a)^3, \qquad -\mathrm{f}(b) = p(c - b)^3, \text{ i.e. } \mathrm{f}(b) = p(b - c)^3.$$

Since $\mathrm{f}(b) > 0$ we have $p \neq 0$ and $b \neq c$, so we may divide:

$$\frac{\mathrm{f}(a)}{\mathrm{f}(b)} = \left(\frac{c - a}{b - c}\right)^3.$$

Every real number has exactly one real cube root, so taking cube roots is reversible and

$$\frac{c - a}{b - c} = \left(\frac{\mathrm{f}(a)}{\mathrm{f}(b)}\right)^{1/3},$$

as required. Writing k for the right-hand side ($k > 0$), $c - a = k(b - c)$, so $c(1 + k) = a + kb$ and

$$c = \frac{a + kb}{1 + k} = \frac{a\,\mathrm{f}(b)^{1/3} + b\,\mathrm{f}(a)^{1/3}}{\mathrm{f}(a)^{1/3} + \mathrm{f}(b)^{1/3}},$$

multiplying top and bottom by $\mathrm{f}(b)^{1/3}$.

? **Ask yourself** Where have we seen this formula before?

It is exactly the formula of part (i) with f replaced by $\mathrm{f}^{1/3}$, which is also defined and positive for $x > 0$. So c is the value of m for the function $\mathrm{f}^{1/3}$. Apply part (iii) with $\mathrm{g}_1 = \mathrm{f}$ and $\mathrm{g}_2 = \mathrm{f}^{1/3}$:

$$\frac{\mathrm{g}_1(x)}{\mathrm{g}_2(x)} = \mathrm{f}(x)^{2/3}.$$

If f is decreasing and positive, then so is $\mathrm{f}^{2/3}$, since $t \mapsto t^{2/3}$ is increasing for $t > 0$. Part (iii) therefore gives $M_1 > M_2$, where $M_1 = m$ (for f itself) and $M_2 = c$. Hence $c < m$.

Checks

▶ Take $a = 1$, $b = 4$. The three means are $2.5 > 2 > 1.6$, and $\mathrm{f} = \sqrt{x}$ gives $m = (1 \cdot 2 + 4 \cdot 1)/3 = 2 = \sqrt{4}$.

- With $\mathrm{f}(x) = 1/x$, $a = 1$, $b = 4$: $m = \frac{1/4+4}{5/4} = 3.4$, while $k = 4^{1/3} \approx 1.587$ gives $c \approx 2.84 < 3.4$, as part (iv) predicts. Both m and c lie between a and b, as weighted averages must.
- Setting $n = 0$ in part (ii) gives $\mathrm{f} = 1$ and $m = \frac{a+b}{2}$; setting $n = -1$ gives $\mathrm{f} = x$ and $m = \frac{2ab}{a+b}$, consistent with part (iii).

These, and random tests of every result over many choices of a, b and f, were confirmed by computer.

Common slips

- Forgetting that B is at height $-\mathrm{f}(b)$, which gives a denominator $\mathrm{f}(b) - \mathrm{f}(a)$ and a point that may not lie between a and b.
- In part (iii), trying to compare M_1 and M_2 by cross-multiplying four function values at once instead of reducing to the single ratio r.
- In part (iv), taking $\mathrm{g}_1 = \mathrm{f}^{1/3}$, $\mathrm{g}_2 = \mathrm{f}$ the wrong way round: their ratio $\mathrm{f}^{-2/3}$ is *increasing*, so part (iii) does not apply in that order.

Beyond STEP

Every m here is a weighted mean $\lambda a + (1-\lambda)b$, and part (iii) is a neat way of ordering the whole family of power means $\left(a^{n+1} + b^{n+1}\right) / \left(a^n + b^n\right)$ (the Lehmer means), which increase with n. Inequalities between means, and the convexity arguments behind them, are a staple of first-year analysis. Abbott's *Understanding Analysis* and Siklos's *Advanced Problems in Mathematics* are good places to see more of this style of argument.

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